COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 1 : Session – 12 : SLO – 2 | Logic gates (combined)
Name: JATANPREET SINGH VIRDI JATANPREET SINGH VIRDI | Reg. No: RA2511003011374 | Solved Worksheet

II. Multiple Choice Questions

• 1. b — AND
• 2. b — Y = A■
• 3. c — NAND
• 4. b — Inputs are the same
• 5. b — Digital comparator

I. Match the Following

• 1. AND gate logic symbol — b) Y = A■B
• 2. OR gate output when both inputs are 0 — 0 (LOW)
• 3. NOT gate Boolean expression — c) Y = A■
• 4. NAND gate function — d) Universal gate for implementing AND, OR, NOT
• 5. EX-NOR gate application — e) Digital comparators and equality checking

III. Activity Questions

• 1. AND/OR/NOT circuit example: F = (A·B) + C■ — AND of A,B ORed with NOT of C. Truth table: F=1 whenever A=B=1 or C=0. Symbols drawn as D-shape (AND), curved (OR), triangle+bubble (NOT).
• 2. NAND: Y = (A·B)' — output 0 only when both inputs are 1. Universal implementations: NOT = A NAND A; AND = (A NAND B) NAND (A NAND B); OR = (A NAND A) NAND (B NAND B).
• 3. EX-NOR: Y = (A⊕B)■ = AB + A■B■ — output 1 when inputs are equal. Truth table: 00→1, 01→0, 10→0, 11→1. Applications: digital comparators, equality checking, parity checking.